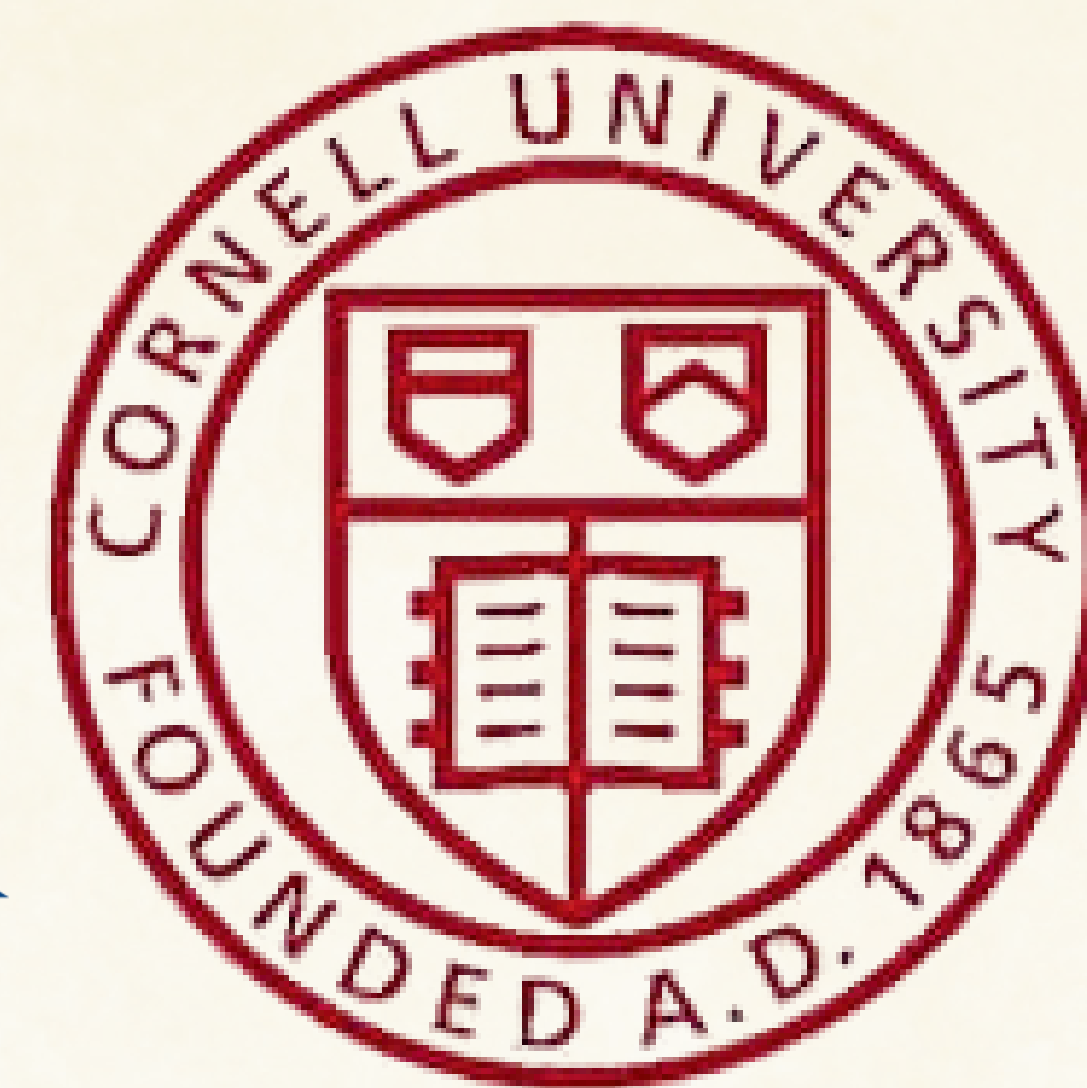




BER - BLOSSOM END ROTTING

RELATION TO SOIL MOISTURE

Investigating the Effects of Soil Moisture Variability on Tomato Fruit Quality and Blossom-End Rot Using an Automated Monitoring and Irrigation System



1. INTRODUCTION

Blossom-end rot (BER) is a common physiological disorder in tomatoes that significantly reduces fruit quality and marketability (Topcu et al., 2022). While BER is often associated with calcium deficiency, research suggests that fluctuations in soil moisture can interfere with calcium transport within the plant (Nicolas et al., 2023). Understanding how soil moisture variability affects BER and overall fruit quality may help improve irrigation practices and crop productivity.



2. QUESTION/HYPOTHESIS

- How does soil moisture variability affect tomato fruit quality and blossom-end rot incidence?
- Tomato plants exposed to inconsistent soil moisture will exhibit reduced fruit quality and increased blossom-end rot incidence compared to plants receiving consistent irrigation.



3. METHODOLOGY



- Twenty tomato plants were divided into two treatment groups:



- Control Group:** Consistent watering schedule (10min every day at 8:00 p.m.)



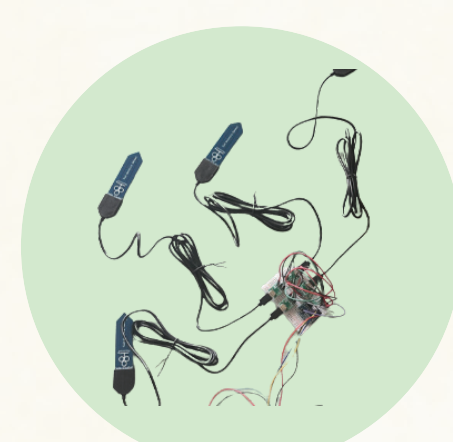
- Experimental Group:** Variable watering schedule (30min every 3 days at 8:00 p.m.)



- Tomato Varieties:** Cherokee Purple, Better Boy, Beefsteak, Roma, Rutgers. Tomatoes are planted in a 5x4 pattern. Each column contains all five varieties to reduce the effect of variety as a confounding variable.

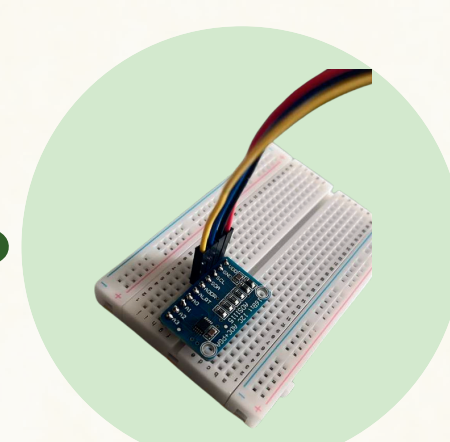


4. ENGINEERING SYSTEM



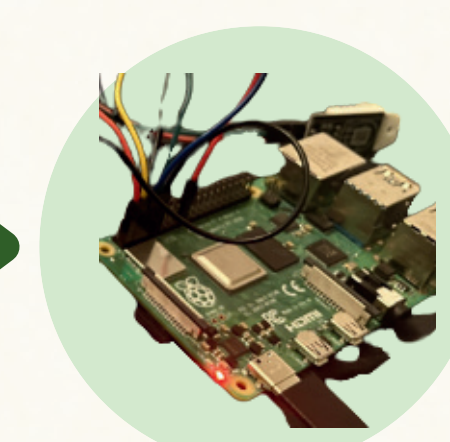
Environmental Sensors

- 4 Moisture Sensors
- DHT22 Temperature / Humidity Sensor



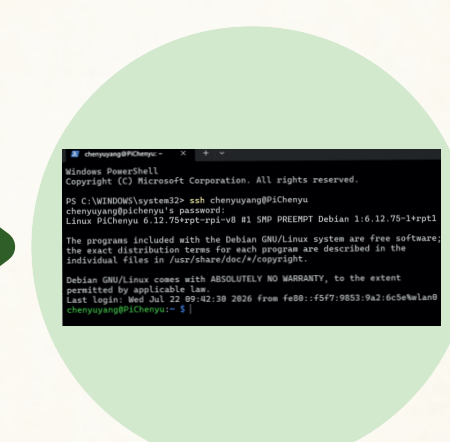
Data Acquisition

- ADS1115 converts analog data into digital values



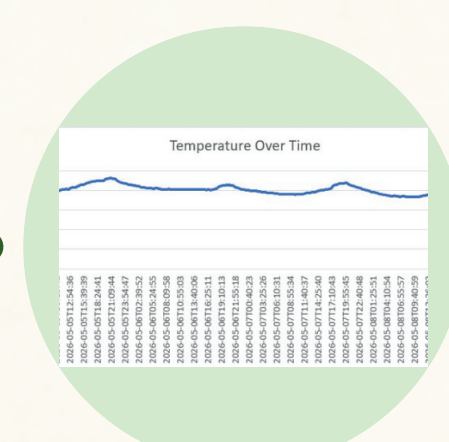
Embedded Computing

- Raspberry Pi
- Automated continuous sampling



Software

- CSV Data Storage
- Timestamped Measurements



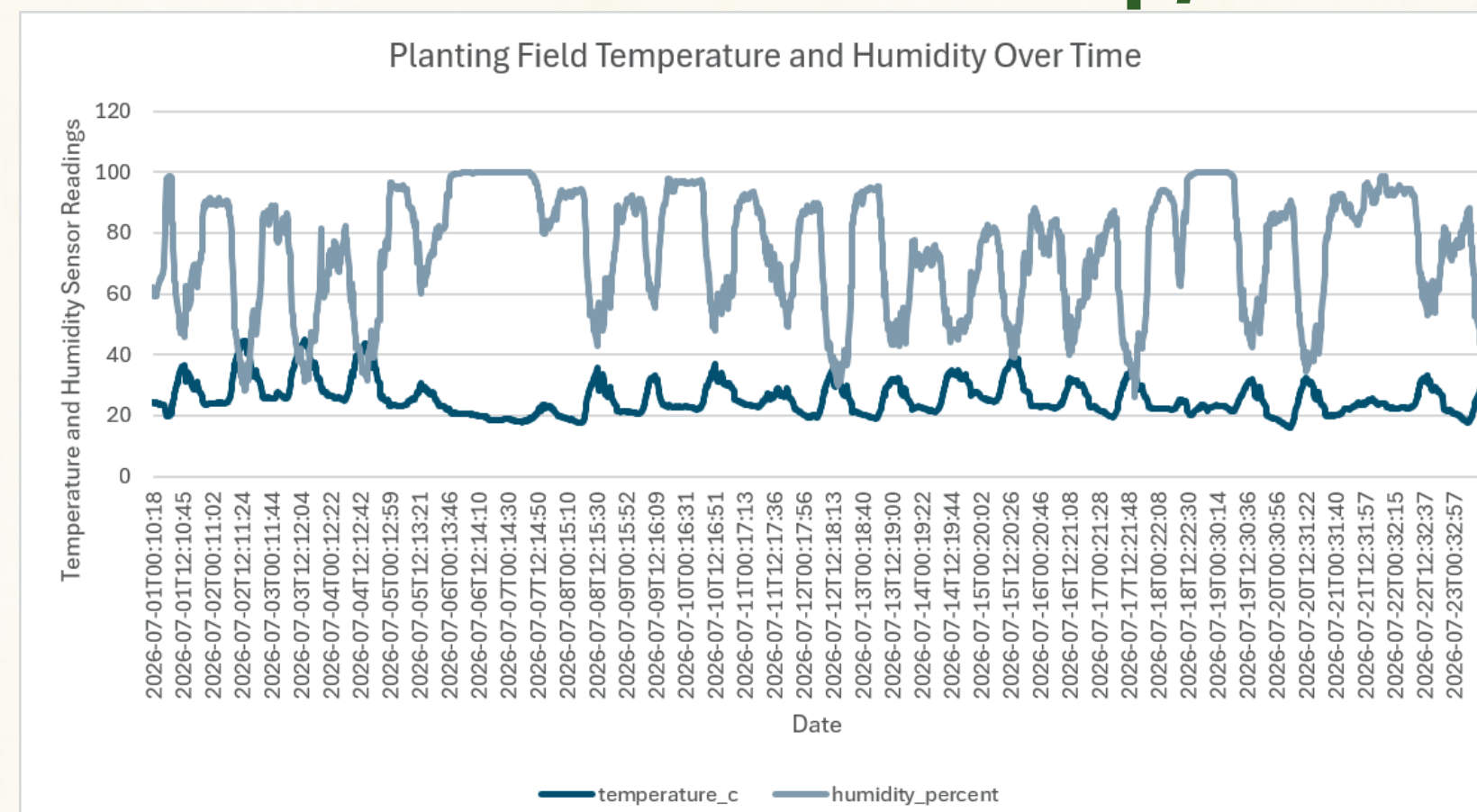
Research Output

- Soil Moisture Trends
- Temperature / Humidity Analysis

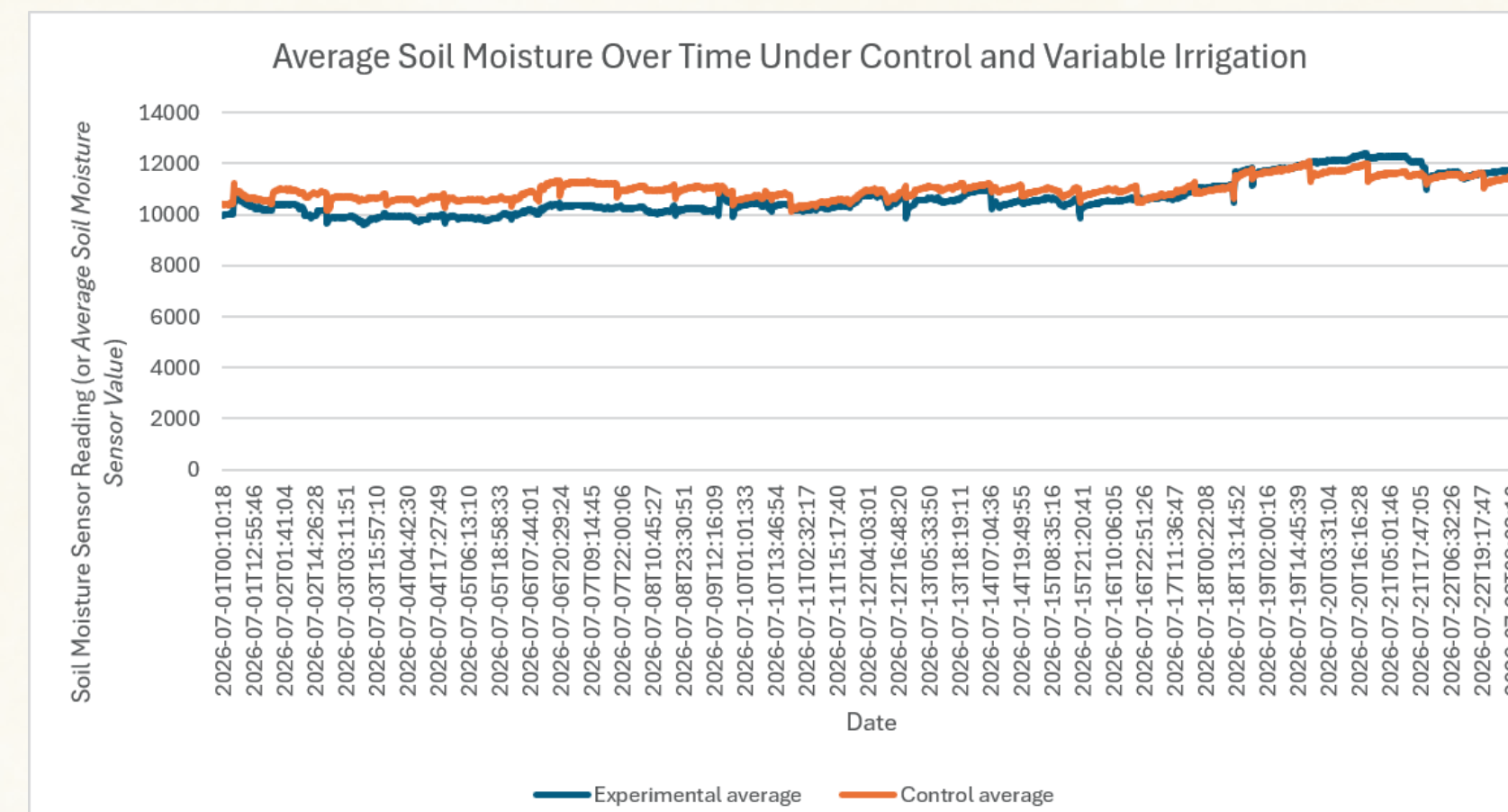


5. PRELIMINARY RESULTS

5.1 Environmental Data - Temp/Humidity



5.2 Environmental Data - Soil Moisture



5.3 Monitoring Metrics

Metric	Current Status
Monitoring Duration	37 Days
Sample Interval	15 Minutes
Total Measurements	34610
Irrigation Treatments	Began July 3

5.4 Observations



5.5 Key Findings

- One blossom-end rot (BER) case was observed in the control treatment; no BER symptoms have been observed in the experimental treatment to date.
- Because most fruit are still developing, additional monitoring is needed before treatment effects on BER incidence can be evaluated.
- The only ripe fruit is the one that shows symptoms consistent with BER.



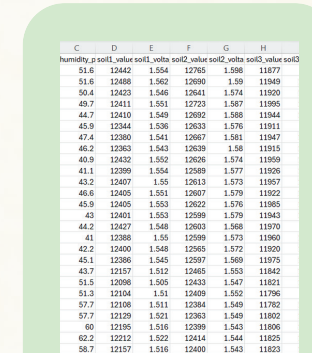
6. FUTURE WORK



Continue Monitoring



Optimize Irrigation



Expand Data Analysis



7. CONCLUSION

- Successfully developed an automated Raspberry Pi-based environmental monitoring system.
- Collected over 34,000 environmental and soil moisture measurements during a 37-day monitoring period.
- Fruit development is ongoing; one BER case has been observed in the control treatment.
- Continued monitoring will determine whether irrigation-induced soil moisture variability influences blossom-end rot incidence.



REFERENCES

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- Topcu, Y., Nambesani, S. U., & Van Der Knaap, E. (2022). Blossom-end rot: a century-old problem in tomato (*Solanum lycopersicum* L.) and other vegetables. *Molecular Horticulture*, 2(1), 1. <https://doi.org/10.1186/s43897-021-00022-9>



ACKNOWLEDGEMENT

I would like to thank my mentor, Researcher Yao Chen at Boyce Thompson Institute for direction, guidance, and support throughout the development of this project. I would also like to thank Cornell and BTI for this opportunity.

Thank you for your time and feedback.



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